

Shah Md. Tasrif Rahman

* Encoding: UTF-8.

* Messed this one up on the first time of submission, now Imma redo allat again ah.

* Task-1 [Data importing/loading]:

```
GET DATA
  /TYPE=XLSX
  /FILE='E:\Statistical
Softwares\SPSS\CUSPC-SPSS\Assignments\Assignment-1\week1_data.xlsx'
  /SHEET=name 'Sheet1'
  /CELLRANGE=FULL
  /READNAMES=ON
  /DATATYPEMIN PERCENTAGE=95.0
  /HIDDEN IGNORE=YES.
EXECUTE.
DATASET NAME DataSet1 WINDOW=FRONT.
```

* Task-2 [Data saving]:

```
SAVE OUTFILE='E:\Statistical
Softwares\SPSS\CUSPC-SPSS\Assignments\Assignment-1\assignment1_Shah_Md._Tasrif_Rahm
an\assignment1_data_Shah_Md._Tasrif_Rahman.sav'
/COMPRESSED.
```

* Task-3 [Defining variables]:

```
DATASET ACTIVATE DataSet1.
* Define Variable Properties.
*student_id.
VARIABLE LEVEL student_id(NOMINAL).
VARIABLE LABELS student_id 'Student identification number'.
```

```
*gender.
VARIABLE LABELS gender 'Gender of respondent'.
VALUE LABELS gender
  '1      ' 'Male'
  '2      ' 'Female'.
```

* This following part is redundant since in the given dataset, the "faculty" column already has the names of the faculties, instead of numbers representing them.

```
*faculty.
*VARIABLE LABELS faculty 'Faculty of enrolment'.
*VALUE LABELS faculty
  'Science'      '1'
  'Business'     '2'
  'Arts'         '3'
  'Engineering'  '4'.
```

```
*year_of_study.
VARIABLE LEVEL year_of_study(SCALE).
VARIABLE LABELS year_of_study 'Current academic year'.
```

```
*study_hours_week.
VARIABLE LABELS study_hours_week 'Self-reported study hours per week'.
```

```
*sleep_hours.
VARIABLE LABELS sleep_hours 'Average hours of sleep per night'.
```

```
*attendance_pct.
VARIABLE LABELS attendance_pct 'Class attendance (percentage)'.
```

```
*stress1.
```

```

VARIABLE LEVEL stress1(ORDINAL).
VARIABLE LABELS stress1 'I feel overwhelmed by coursework'.
VALUE LABELS stress1
  1 'Strongly disagree'
  2 'Disagree'
  3 'Neutral'
  4 'Agree'
  5 'Strongly agree'.

*stress2.
VARIABLE LEVEL stress2(ORDINAL).
VARIABLE LABELS stress2 'I worry about exams'.
VALUE LABELS stress2
  1 'Strongly disagree'
  2 'Disagree'
  3 'Neutral'
  4 'Agree'
  5 'Strongly agree'.

*stress3.
VARIABLE LEVEL stress3(ORDINAL).
VARIABLE LABELS stress3 'I find it hard to relax'.
VALUE LABELS stress3
  1 'Strongly disagree'
  2 'Disagree'
  3 'Neutral'
  4 'Agree'
  5 'Strongly agree'.

*stress4.
VARIABLE LEVEL stress4(ORDINAL).
VARIABLE LABELS stress4 'I feel tense about deadlines'.
VALUE LABELS stress4
  1 'Strongly disagree'
  2 'Disagree'
  3 'Neutral'
  4 'Agree'
  5 'Strongly agree'.

*stress5_calm.
VARIABLE LEVEL stress5_calm(ORDINAL).
VARIABLE LABELS stress5_calm 'I feel calm and in control (REVERSE-WORDED)'.
VALUE LABELS stress5_calm
  1 'Strongly disagree'
  2 'Disagree'
  3 'Neutral'
  4 'Agree'
  5 'Strongly agree'.

*gpa.
VARIABLE LABELS gpa 'Grade point average (0.00 - 4.00)'.
EXECUTE.

* Task-4 [Recoding gender column and checking consistency]:

DATASET ACTIVATE DataSet1.
RECODE gender ('1'='Male') ('Male'='Male') ('male'='Male') ('M'='Male') ('2'='Female')
  ('Female'='Female') ('female'='Female') ('F'='Female').
EXECUTE.

DATASET ACTIVATE DataSet1.
RECODE gender ('Male'='1') ('Female'='2').
EXECUTE.
VARIABLE LEVEL gender(SCALE).
VARIABLE LABELS gender 'Gender of respondent'.
VALUE LABELS gender
  '1      ' 'Male'
  '2      ' 'Female'.

```

```
EXECUTE.
```

```
FREQUENCIES VARIABLES=gender  
  /STATISTICS=RANGE MINIMUM MAXIMUM STDDEV MEAN MEDIAN  
  /FORMAT=NOTABLE  
  /ORDER=ANALYSIS.
```

Frequencies

```
[DataSet1] E:\Statistical  
Softwares\SPSS\CUSPC-SPSS\Assignments\Assignment-1\assignment1_Shah_Md._Tasrif_Rahm  
an\assignment1_data_Shah_Md._Tasrif_Rahman.sav
```

Statistics

Gender of respondent

N	Valid	302
	Missing	0

```
FREQUENCIES VARIABLES=year_of_study  
  /STATISTICS=RANGE MINIMUM MAXIMUM STDDEV MEAN MEDIAN  
  /FORMAT=NOTABLE  
  /ORDER=ANALYSIS.
```

Frequencies

Statistics

Current academic year

N	Valid	302
	Missing	0
Mean		2.49
Median		3.00
Std. Deviation		1.065
Range		3
Minimum		1
Maximum		4

```

FREQUENCIES VARIABLES=study_hours_week
  /STATISTICS=RANGE MINIMUM MAXIMUM STDDEV MEAN MEDIAN
  /FORMAT=NOTABLE
  /ORDER=ANALYSIS.

```

Frequencies

Statistics

Self-reported study hours per
week

N	Valid	289
	Missing	13
Mean		13.648
Median		13.800
Std. Deviation		4.7901
Range		25.2
Minimum		1.0
Maximum		26.2

```

FREQUENCIES VARIABLES=sleep_hours
  /STATISTICS=RANGE MINIMUM MAXIMUM STDDEV MEAN MEDIAN
  /FORMAT=NOTABLE
  /ORDER=ANALYSIS.

```

Frequencies

Statistics

Average hours of sleep per night

N	Valid	284
	Missing	18
Mean		7.689
Median		6.800
Std. Deviation		9.5156
Range		95.5
Minimum		3.5
Maximum		99.0

```
FREQUENCIES VARIABLES=attendance_pct
  /STATISTICS=RANGE MINIMUM MAXIMUM STDDEV MEAN MEDIAN
  /FORMAT=NOTABLE
  /ORDER=ANALYSIS.
```

Frequencies

Statistics

Class attendance (percentage)

N	Valid	287
	Missing	15
Mean		78.616
Median		77.800
Std. Deviation		18.6163
Range		207.9
Minimum		42.1
Maximum		250.0

```
FREQUENCIES VARIABLES=gpa
  /STATISTICS=RANGE MINIMUM MAXIMUM STDDEV MEAN MEDIAN
  /FORMAT=NOTABLE
  /ORDER=ANALYSIS.
```

Frequencies

Statistics

Grade point average (0.00 - 4.00)

N	Valid	302
	Missing	0
Mean		3.6625
Median		3.8000
Std. Deviation		.38073
Range		1.76
Minimum		2.24
Maximum		4.00

* Task-5 [Fixing missing values]:

```
MISSING VALUES sleep_hours (99)
EXECUTE.
```

Error # 701 in column 1. Text: EXECUTE
An undefined variable name, or a scratch or system variable was specified in a variable list which accepts only standard variables. Check spelling and verify the existence of this variable.
Execution of this command stops.

Error # 4816. Command name: MISSING VALUES
The variable list on the MISSING VALUES command was incorrect.

Error # 4810 in column 1024. Text: (End of Command)
The parenthesized list of missing values was omitted.
MISSING VALUES attendance_pct (250)
EXECUTE.

Error # 701 in column 1. Text: EXECUTE
An undefined variable name, or a scratch or system variable was specified in a variable list which accepts only standard variables. Check spelling and verify the existence of this variable.
Execution of this command stops.

Error # 4816. Command name: MISSING VALUES
The variable list on the MISSING VALUES command was incorrect.

Error # 4810 in column 1024. Text: (End of Command)
The parenthesized list of missing values was omitted.

* Task-6 [Finding and removing duplicate observations]:

```
* Identify Duplicate Cases.
SORT CASES BY student_id(A).
MATCH FILES
  /FILE=*
  /BY student_id
  /FIRST=PrimaryFirst
  /LAST=PrimaryLast_sorted_by_student_id.
DO IF (PrimaryFirst).
COMPUTE MatchSequence=1-PrimaryLast_sorted_by_student_id.
```

```

ELSE.
COMPUTE MatchSequence=MatchSequence+1.
END IF.
LEAVE MatchSequence.
FORMATS MatchSequence (f7).
COMPUTE InDupGrp=MatchSequence>0.
SORT CASES InDupGrp(D).
MATCH FILES
  /FILE=*
  /DROP=PrimaryFirst InDupGrp MatchSequence.
VARIABLE LABELS PrimaryLast_sorted_by_student_id 'Indicator of each last matching case
as Primary'.
VALUE LABELS PrimaryLast_sorted_by_student_id 0 'Duplicate Case' 1 'Primary Case'.
VARIABLE LEVEL PrimaryLast_sorted_by_student_id (ORDINAL).
FREQUENCIES VARIABLES=PrimaryLast_sorted_by_student_id.

```

Frequencies

Statistics

Indicator of each last matching
case as Primary

N	Valid	302
	Missing	0

Indicator of each last matching case as Primary

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Duplicate Case	2	.7	.7	.7
	Primary Case	300	99.3	99.3	100.0
	Total	302	100.0	100.0	

EXECUTE.

* Removing duplicates permanently:

```

SELECT IF (PrimaryLast_sorted_by_student_id = 1).
EXECUTE.

```

*There were 2 duplicate cases out of 302 cases, 300 primary cases remain after deduplication.

* Task-7 [Reverse coding and adding stress_total variable]:

```

COMPUTE stress5_r=6-stress5_calm.
EXECUTE.

```

```

COMPUTE stress_total=stress1 + stress2 + stress3 + stress4 + stress5_r.

```

EXECUTE.

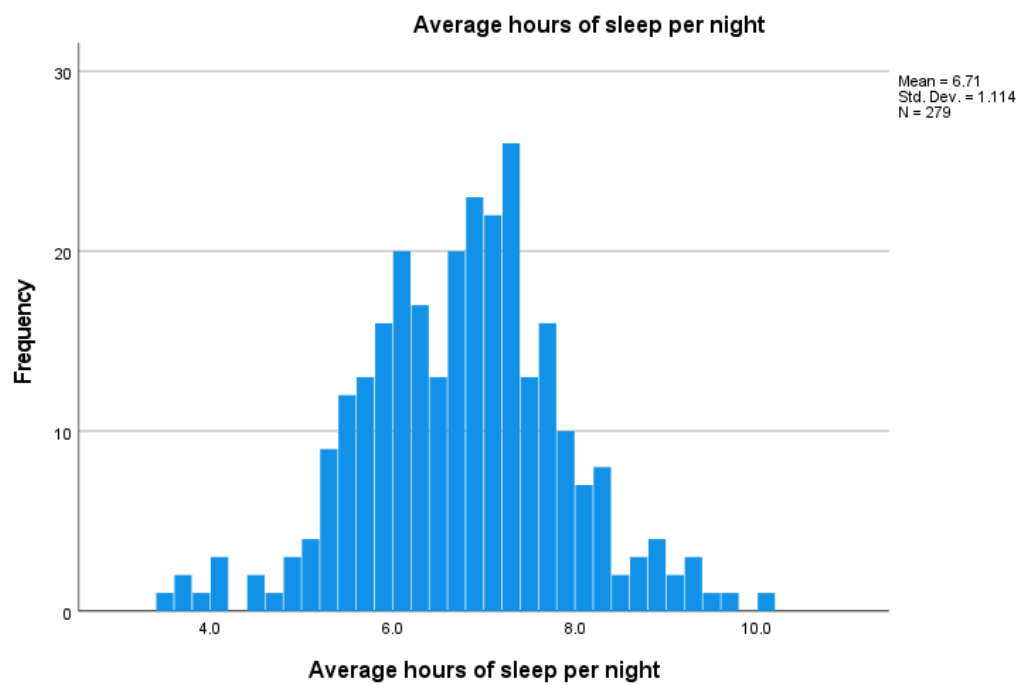
* Task-8 [Checking summary and drawing histogram]:

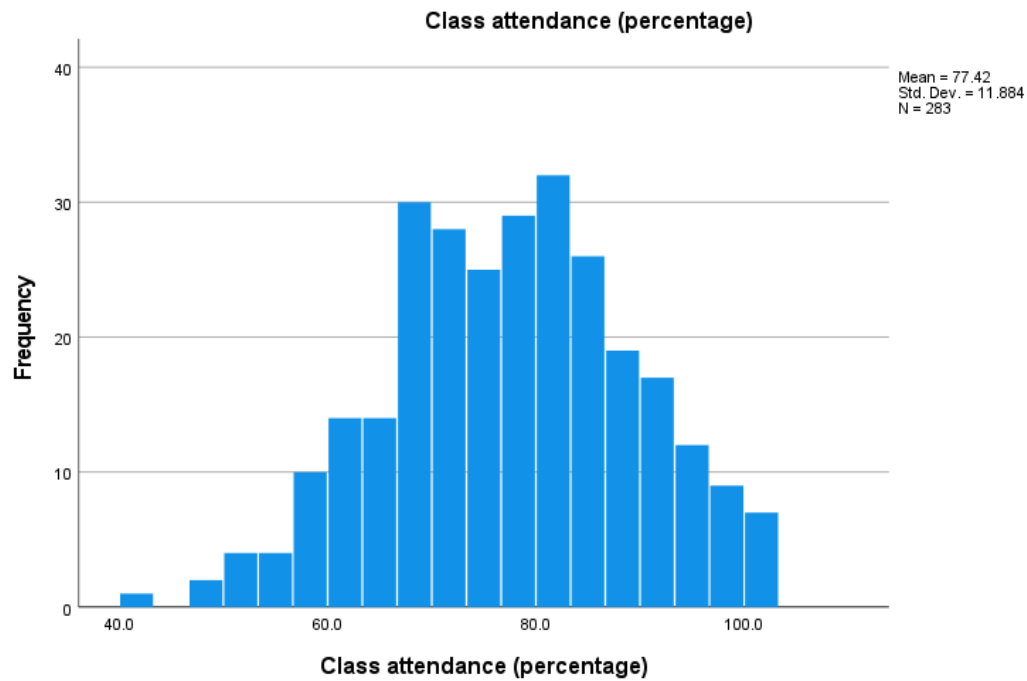
```
FREQUENCIES VARIABLES=study_hours_week sleep_hours attendance_pct
/FORMAT=NOTABLE
/STATISTICS=RANGE MINIMUM MAXIMUM MEAN MEDIAN
/HISTOGRAM
/ORDER=ANALYSIS.
```

Frequencies

Statistics				
		Self-reported study hours per week	Average hours of sleep per night	Class attendance (percentage)
N	Valid	288	279	283
	Missing	12	21	17
Mean		13.631	6.714	77.419
Median		13.750	6.800	77.600
Range		25.2	6.5	57.9
Minimum		1.0	3.5	42.1
Maximum		26.2	10.0	100.0

Histogram





* Task-9 [Checking frequency table and drawing bar chart]:

```
FREQUENCIES VARIABLES=faculty year_of_study
  /BARCHART FREQ
  /ORDER=ANALYSIS.
```

Frequencies

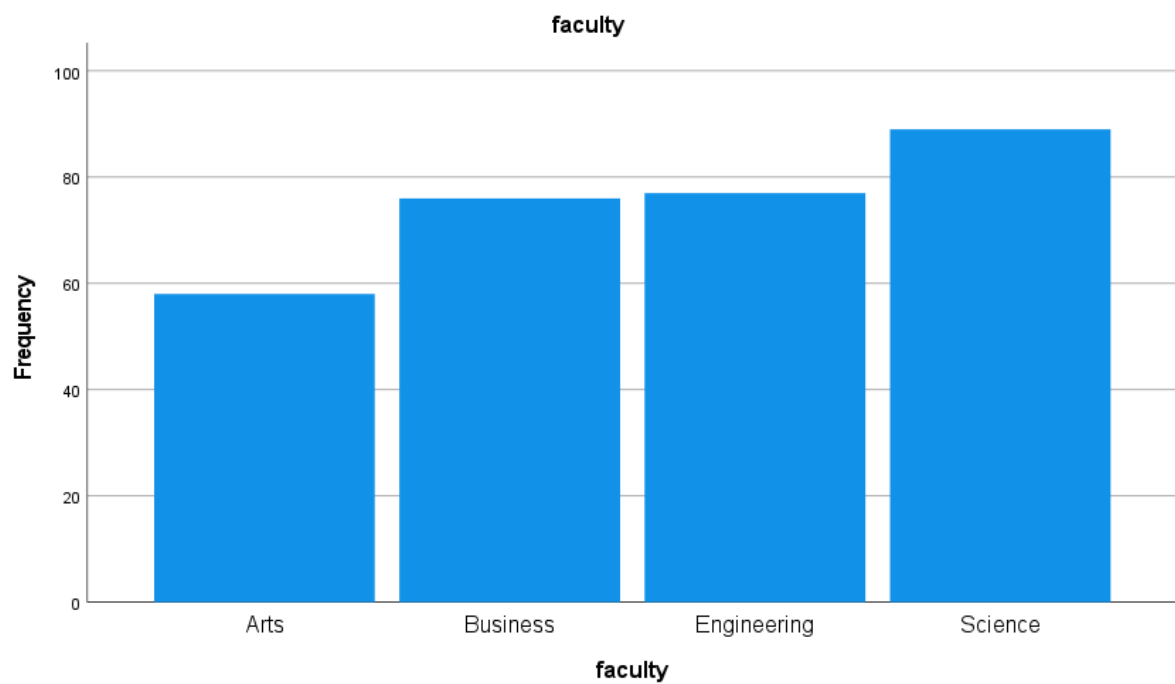
		Statistics	
			Current
		faculty	academic year
N	Valid	300	300
	Missing	0	0

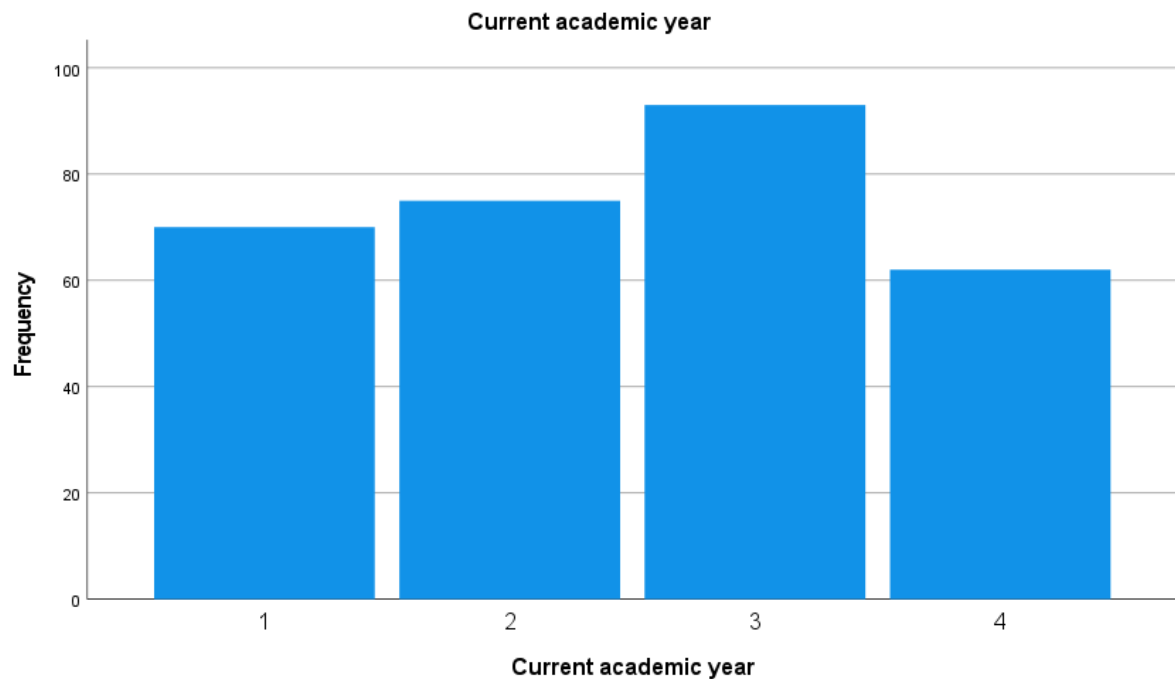
Frequency Table

		faculty			
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Arts	58	19.3	19.3	19.3
	Business	76	25.3	25.3	44.7
	Engineering	77	25.7	25.7	70.3
	Science	89	29.7	29.7	100.0
	Total	300	100.0	100.0	

		Current academic year			
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	1	70	23.3	23.3	23.3
	2	75	25.0	25.0	48.3
	3	93	31.0	31.0	79.3
	4	62	20.7	20.7	100.0
	Total	300	100.0	100.0	

Bar Chart





```
DATASET ACTIVATE DataSet1.
```

```
SAVE OUTFILE='E:\Statistical '+'
```

```
'Softwares\SPSS\CUSPC-SPSS\Assignments\Assignment-1\assignment1_Shah_Md._Tasrif_Rahman\assignme'+
'nt1_data_Shah_Md._Tasrif_Rahman.sav'
/COMPRESSED.
```

```
* Task-10 [Exporting output]:
```

```
OUTPUT SAVE NAME=Document1
OUTFILE='E:\Statistical '+'
```

```
'Softwares\SPSS\CUSPC-SPSS\Assignments\Assignment-1\assignment1_Shah_Md._Tasrif_Rahman\assignme'+
'nt1_Shah_Md._Tasrif_Rahman_Output.spv'
LOCK=NO.
```

```
* Export Output.
```

```
OUTPUT EXPORT
/CONTENTS EXPORT=VISIBLE LAYERS=PRINTSETTING MODELVIEWS=PRINTSETTING
/DOC DOCUMENTFILE='E:\Statistical '+'
```

```
'Softwares\SPSS\CUSPC-SPSS\Assignments\Assignment-1\assignment1_Shah_Md._Tasrif_Rahman\assignme'+
'nt1_data_Shah_Md._Tasrif_Rahman.doc'
NOTESCAPTIONS=YES WIDETABLES=SHRINK PAGEBREAKS=YES
PAGESIZE=INCHES(8.267000000000001, 11.69) TOPMARGIN=INCHES(1.0)
BOTTOMMARGIN=INCHES(1.0)
```

Appendix:

* Encoding: UTF-8.

* Messed this one up on the first time of submission, now Imma redo allat again ah.

* Task-1 [Data importing/loading]:

```
GET DATA
  /TYPE=XLSX
  /FILE='E:\Statistical
Softwares\SPSS\CUSPC-SPSS\Assignments\Assignment-1\week1_data.xlsx'
  /SHEET=name 'Sheet1'
  /CELLRANGE=FULL
  /READNAMES=ON
  /DATATYPEMIN PERCENTAGE=95.0
  /HIDDEN IGNORE=YES.
EXECUTE.
DATASET NAME DataSet1 WINDOW=FRONT.
```

* Task-2 [Data saving]:

```
SAVE OUTFILE='E:\Statistical
Softwares\SPSS\CUSPC-SPSS\Assignments\Assignment-1\assignment1_Shah_Md._Tas
rif_Rahman\assignment1_data_Shah_Md._Tasrif_Rahman.sav'
/COMPRESSED.
```

* Task-3 [Defining variables]:

```
DATASET ACTIVATE DataSet1.
* Define Variable Properties.
*student_id.
VARIABLE LEVEL student_id(NOMINAL).
VARIABLE LABELS student_id 'Student identification number'.
```

```
*gender.
VARIABLE LABELS gender 'Gender of respondent'.
VALUE LABELS gender
  '1      ' 'Male'
  '2      ' 'Female'.
```

* This following part is redundant since in the given dataset, the "faculty" column already has the names of the faculties, instead of numbers representing them.

```
*faculty.
*VARIABLE LABELS faculty 'Faculty of enrolment'.
*VALUE LABELS faculty
  'Science'      '1'
  'Business'     '2'
  'Arts'         '3'
  'Engineering'  '4'.
```

```
*year_of_study.
VARIABLE LEVEL year_of_study(SCALE).
VARIABLE LABELS year_of_study 'Current academic year'.
```

```
*study_hours_week.
VARIABLE LABELS study_hours_week 'Self-reported study hours per week'.
```

```

*sleep_hours.
VARIABLE LABELS  sleep_hours 'Average hours of sleep per night'.

*attendance_pct.
VARIABLE LABELS  attendance_pct 'Class attendance (percentage)'.

*stress1.
VARIABLE LEVEL  stress1(ORDINAL).
VARIABLE LABELS  stress1 'I feel overwhelmed by coursework'.
VALUE LABELS stress1
  1 'Strongly disagree'
  2 'Disagree'
  3 'Neutral'
  4 'Agree'
  5 'Strongly agree'.

*stress2.
VARIABLE LEVEL  stress2(ORDINAL).
VARIABLE LABELS  stress2 'I worry about exams'.
VALUE LABELS stress2
  1 'Strongly disagree'
  2 'Disagree'
  3 'Neutral'
  4 'Agree'
  5 'Strongly agree'.

*stress3.
VARIABLE LEVEL  stress3(ORDINAL).
VARIABLE LABELS  stress3 'I find it hard to relax'.
VALUE LABELS stress3
  1 'Strongly disagree'
  2 'Disagree'
  3 'Neutral'
  4 'Agree'
  5 'Strongly agree'.

*stress4.
VARIABLE LEVEL  stress4(ORDINAL).
VARIABLE LABELS  stress4 'I feel tense about deadlines'.
VALUE LABELS stress4
  1 'Strongly disagree'
  2 'Disagree'
  3 'Neutral'
  4 'Agree'
  5 'Strongly agree'.

*stress5_calm.
VARIABLE LEVEL  stress5_calm(ORDINAL).
VARIABLE LABELS  stress5_calm 'I feel calm and in control (REVERSE-WORDED)'.
VALUE LABELS stress5_calm
  1 'Strongly disagree'
  2 'Disagree'
  3 'Neutral'
  4 'Agree'
  5 'Strongly agree'.

*gpa.
VARIABLE LABELS  gpa 'Grade point average (0.00 - 4.00)'.
EXECUTE.

* Task-4 [Recoding gender column and checking consistency]:

```

```

DATASET ACTIVATE DataSet1.
RECODE gender ('1'='Male') ('Male'='Male') ('male'='Male') ('M'='Male')
('2'='Female')
('Female'='Female') ('female'='Female') ('F'='Female').
EXECUTE.

```

```

DATASET ACTIVATE DataSet1.
RECODE gender ('Male'='1') ('Female'='2').
EXECUTE.
VARIABLE LEVEL gender(SCALE).
VARIABLE LABELS gender 'Gender of respondent'.
VALUE LABELS gender
'1      ' 'Male'
'2      ' 'Female'.
EXECUTE.

```

```

FREQUENCIES VARIABLES=gender
/STATISTICS=RANGE MINIMUM MAXIMUM STDDEV MEAN MEDIAN
/FORMAT=NOTABLE
/ORDER=ANALYSIS.

```

```

FREQUENCIES VARIABLES=year_of_study
/STATISTICS=RANGE MINIMUM MAXIMUM STDDEV MEAN MEDIAN
/FORMAT=NOTABLE
/ORDER=ANALYSIS.

```

```

FREQUENCIES VARIABLES=study_hours_week
/STATISTICS=RANGE MINIMUM MAXIMUM STDDEV MEAN MEDIAN
/FORMAT=NOTABLE
/ORDER=ANALYSIS.

```

```

FREQUENCIES VARIABLES=sleep_hours
/STATISTICS=RANGE MINIMUM MAXIMUM STDDEV MEAN MEDIAN
/FORMAT=NOTABLE
/ORDER=ANALYSIS.

```

```

FREQUENCIES VARIABLES=attendance_pct
/STATISTICS=RANGE MINIMUM MAXIMUM STDDEV MEAN MEDIAN
/FORMAT=NOTABLE
/ORDER=ANALYSIS.

```

```

FREQUENCIES VARIABLES=gpa
/STATISTICS=RANGE MINIMUM MAXIMUM STDDEV MEAN MEDIAN
/FORMAT=NOTABLE
/ORDER=ANALYSIS.

```

* Task-5 [Fixing missing values]:

```

MISSING VALUES sleep_hours (99)
EXECUTE.
MISSING VALUES attendance_pct (250)
EXECUTE.

```

* Task-6 [Finding and removing duplicate observations]:

```

* Identify Duplicate Cases.
SORT CASES BY student_id(A).
MATCH FILES
/FILE=*
/BY student_id

```

```

    /FIRST=PrimaryFirst
    /LAST=PrimaryLast_sorted_by_student_id.
DO IF (PrimaryFirst).
COMPUTE MatchSequence=1-PrimaryLast_sorted_by_student_id.
ELSE.
COMPUTE MatchSequence=MatchSequence+1.
END IF.
LEAVE MatchSequence.
FORMATS MatchSequence (f7).
COMPUTE InDupGrp=MatchSequence>0.
SORT CASES InDupGrp(D).
MATCH FILES
    /FILE=*
    /DROP=PrimaryFirst InDupGrp MatchSequence.
VARIABLE LABELS PrimaryLast_sorted_by_student_id 'Indicator of each last
matching case as Primary'.
VALUE LABELS PrimaryLast_sorted_by_student_id 0 'Duplicate Case' 1 'Primary
Case'.
VARIABLE LEVEL PrimaryLast_sorted_by_student_id (ORDINAL).
FREQUENCIES VARIABLES=PrimaryLast_sorted_by_student_id.
EXECUTE.

* Removing duplicates permanently:

SELECT IF (PrimaryLast_sorted_by_student_id = 1).
EXECUTE.

*There were 2 duplicate cases out of 302 cases, 300 primary cases remain after
deduplication.

* Task-7 [Reverse coding and adding stress_total variable]:

COMPUTE stress5_r=6-stress5_calm.
EXECUTE.

COMPUTE stress_total=stress1 + stress2 + stress3 + stress4 + stress5_r.
EXECUTE.

* Task-8 [Checking summary and drawing histogram]:

FREQUENCIES VARIABLES=study_hours_week sleep_hours attendance_pct
    /FORMAT=NOTABLE
    /STATISTICS=RANGE MINIMUM MAXIMUM MEAN MEDIAN
    /HISTOGRAM
    /ORDER=ANALYSIS.

* Task-9 [Checking frequency table and drawing bar chart]:

FREQUENCIES VARIABLES=faculty year_of_study
    /BARCHART FREQ
    /ORDER=ANALYSIS.

DATASET ACTIVATE DataSet1.

SAVE OUTFILE='E:\Statistical '+
'Softwares\SPSS\CUSPC-SPSS\Assignments\Assignment-1\assignment1_Shah_Md._Ta
srif_Rahman\assignme'+
'nt1_data_Shah_Md._Tasrif_Rahman.sav'
/COMPRESSED.

```


* Task-10 [Exporting output]:

OUTPUT SAVE NAME=Document1
OUTFILE='E:\Statistical '+'

'Softwares\SPSS\CUSPC-SPSS\Assignments\Assignment-1\assignment1_Shah_Md._Ta
srif_Rahman\assignme'+
 'nt1_Shah_Md._Tasrif_Rahman_Output.spv'
LOCK=NO.

* Export Output.

OUTPUT EXPORT
 /CONTENTS EXPORT=VISIBLE LAYERS=PRINTSETTING MODELVIEWS=PRINTSETTING
 /DOC DOCUMENTFILE='E:\Statistical '+'

'Softwares\SPSS\CUSPC-SPSS\Assignments\Assignment-1\assignment1_Shah_Md._Ta
srif_Rahman\assignme'+
 'nt1_data_Shah_Md._Tasrif_Rahman.doc'
 NOTESCAPTIONS=YES WIDETABLES=SHRINK PAGEBREAKS=YES
 PAGESIZE=INCHES(8.267000000000001, 11.69) TOPMARGIN=INCHES(1.0)
BOTTOMMARGIN=INCHES(1.0)
 LEFTMARGIN=INCHES(1.0) RIGHTMARGIN=INCHES(1.0000000000000018).